

A Objective

This lab develops the skills to add and manage users and groups with the Linux operating systems using shell commands. Further, the lab will look behind the scenes of user management in Linux by analyzing the content of the corresponding configuration files. In the second part of this lab, as a more real-world example, we will create the users of a small company called *Gotham City*!

B Platform and process

- You will need a Linux-system where you have superuser (root) privileges (with `sudo` or `su`) – employ one of the prepared VMs running e.g. Kali Linux (the privileged user is called `kali` or `junioradmin`)!
- Execute all commands in a terminal window using the standard shell and don't forget to elevate your privileges first using e.g. `sudo su`!

→ Be sure to provide answers to all questions and to document **each and every** shell command line in your lab notes (lab report)!

C Existing users

- (1) A *regular* linux user belongs to many groups.
 - a. Login as user `junioradmin`. Which group(s) are you in the (virtual machine) in? *Hint*: use the `id` command.
 - b. Why are there so many groups?
 - c. What about user `kali` (password: `kali`) in the virtual machine – which groups is he in?

D Creating a new user

- (2) Create a new group called `heroesNN` ! Note: replace `NN` by the last two digits of your login-number!

Which (simple) command line did you use to create the new user group?

- (3) Changes
 - a. Which (text) file in `/etc` received a new entry as a consequence to step (2)?
`/etc/group`
 - b. List the corresponding line from the `group` file using the command `grep`!
Command: `grep heroes28 /etc/group`
Result: `heroes28:x:1002:`
 - c. What is the group `id` of the group `heroesNN`?
`1002`
 - d. Try the following command lines – what is the purpose of each?

```
getent passwd
queries user
database
```

```
getent group
queries group
database
```

d. What else can `getent` show? Tipp: Just use tabulator-key on command-line or check the manual pages.

hosts, services, protocols, networks, etc.

(4) Create a new user with the following specs – use only *one single* shell command line:

- User name: max
- Full name: Max Headroom
- Home directory: /home/max
- Shell: /bin/bash
- “Main” group (= *Login Group*): heroesNN

a. What command line did you use to create the user¹

`sudo useradd -m -c "Max Headroom" -d /home/max -s /bin/bash -g heroes28 max`

b. What (text) system files consequently received new entries - enter the absolute path?

List the corresponding lines from the `passwd` file using the command `grep`!

`/etc/passwd, /etc/shadow`

c. What is the (numerical) user id of user max?

`1002`

d. Is it already possible for the user max to login?

No — password not set yet

(5) Configure a *secure* password for user max!

a. How did you do that?

Used `passwd` command

b. What text file in `/etc` was changed when adding the password? List the corresponding line with `grep`!

Which command did you use?

File: `/etc/shadow` — check with `grep max /etc/shadow`

(6) Try to login with user name max – does it work? (If not, what error message do you receive? See (7))

a. ... with the command `su -username`. What does the command `su` stand for?

Switch user

Does it work?

yes

What does the strange option “-” mean?

¹ note the mnemonic trick for the important options of `useradd`: Getting something done makes charts green ;-)

The "-" loads a **full login environment**

- b. ... via the network using `ssh1: sshusername@localhost` Does it work?

```
player@playground:~$ ssh max@localhost
The authenticity of host 'localhost (:::1)' can't be established.
ED25519 key fingerprint is SHA256:rkX7g8OefSzec8oAGP0MPKiyqS5Kg491T+09ilQohA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
max@localhost's password:
script: cannot open /var/log/ssh-sessions/2026-09-21_15-36-10_max_145.log: Permission denied
Connection to localhost closed.
player@playground:~$ S
```

Authentication works, but the SSH session is closed due to an error at `/var/log/ssh-sessions/`

- c. If a) and/or b) work, try to login by using the graphical user interface (GUI)!

Does it work?

I don't have a GUI

```
player@playground:~$ sudo groupadd heroes28
player@playground:~$ grep heroes28 /etc/group
getent group heroes28
heroes28:x:1002:
heroes28:x:1002:
player@playground:~$ sudo useradd -m -c "Max Headroom" -d /home/max -s /bin/bash -g heroes28 max
player@playground:~$ sudo passwd max
New password:
Retype new password:
passwd: password updated successfully
player@playground:~$ su - max
Password:
max@playground:~$
```

- (7) **Only if (6) did not work (couldn't login):** Check if perhaps the home directory of `max` does not exist at all (*how do you check this?*) or if maybe the password has not been set (*where would you check this?*)

- If the home directory does not exist (you may have missed to pass the correct option to `useradd`), you will need to create it using `mkdir`. The *owner* should be `max` – and `max` should have all permissions to his own directory – does the login work now?
- So what was the cause of the problem?

Does it work?

E Creating multiple users

- (8) Create the following users, groups and directories on the system:

- a. ... with the help of the simple *Shell Script* `addusers.sh` (include it in your lab report – you will need it again later!):

¹ you may need to install & activate an SSH-Server first, you can do this using `sudo apt-get install openssh-server` and `sudo systemctl start ssh`

Hint 1: It is easiest to create the script file with an *editor* (vi, nano, gedit...). For each new user, enter a `useradd` command per line. You can execute the script with `sudo bash scriptfile`. This will start the script with superuser (root)-privileges. In this case you don't need `sudo` in front of each line in your script!

Hint 2: It is a good idea to create both the groups (friends, heroesNN, ...) and the higher-level folders (/home/friends, /home/heroesNN, ...) *first*, before adding the users!

Hint 3: You may want to automate adding passwords by entering them directly in the script file (what is the potential problem here?) – to achieve this, you *cannot* use the `passwd` command, as it expects input directly from the console. Use the command `chpasswd` (which reads from standard input, which in turn can be redirected from a pipe) instead, e.g.: `echo username:password | chpasswd`

Hint 4: It would be smart to make backup copies of all three (important) user management files in `/etc` before creating the groups and users – *how do you do that?*

- b. List of users: Note: replace NN by the last two digits of your login-number!

Username	Full Name	Login Group	Home Directory	Password
adminNN	YourName	heroesNN	/home/heroesNN/adminNN	linux!is!c001!
batman	Bruce Wayne	heroesNN	/home/heroesNN/batman	darkknight
robin	Dick Grayson	friends	/home/friends/robin	nightwing
catwoman	Selina Kyle	friends	/home/friends/catwoman	meowmeow
gordon	James Gordon	police	/home/police/gordon	law-n-order
joker	The Joker	villains	/home/villains/joker	whysoserious
penguin	Oswald Cobblepot	villains	/home/villains/penguin	fish!

- (9) a. Ensure that all users (including their home directories) have been successfully created – you can test this by logging in via the GUI or directly in the terminal using `su -username!`

- b. How secure are above passwords?

Very insecure

- c. Your script to create the users

- d. Write and test a second script file `delusers.sh` that *deletes* all the users, groups and directories again! Insert your script as text here:

```
#!/bin/bash
```

```
# Delete users and their home directories
```

```
userdel -r adminNN
```

```
userdel -r batman
```

```
userdel -r robin
```

```
userdel -r catwoman
```

```
userdel -r gordon
```

```
userdel -r joker
```

```
userdel -r penguin
```

```
# Delete groups
```

```
groupdel heroesNN
```

```
groupdel friends
```

```
groupdel police
```

```
groupdel villains
```

```
# Delete remaining directories
```

```
rm -rf /home/heroes28
```

```
rm -rf /home/friends
```

```
rm -rf /home/police
```

```
rm -rf /home/villains
```

(10) *Bonus ;-):* Who is/was Max Headroom? Where does his name come from?

We wish you lots of success and fun with the exercise!

```

player@playground:~$ sudo cp /etc/passwd /etc/passwd.backup
sudo cp /etc/group /etc/group.backup
sudo cp /etc/shadow /etc/shadow.backup
player@playground:~$ nano addusers.sh
player@playground:~$ sudo bash addusers.sh
player@playground:~$ grep batman /etc/passwd
id batman
su - batman
batman:x:1004:1003:Bruce Wayne:/home/heroesNN/batman:/bin/bash
uid=1004(batman) gid=1003(heroesNN) groups=1003(heroesNN)
Password:
su: Authentication failure
player@playground:~$ grep batman /etc/passwd
id batman
su - batman # Login testen
batman:x:1004:1003:Bruce Wayne:/home/heroesNN/batman:/bin/bash
uid=1004(batman) gid=1003(heroesNN) groups=1003(heroesNN)
Password:
su: Authentication failure
player@playground:~$ sudo bash addusers.sh
groupadd: group 'heroesNN' already exists
groupadd: group 'friends' already exists
groupadd: group 'police' already exists
groupadd: group 'villains' already exists
useradd: user 'admin28' already exists
useradd: user 'batman' already exists
useradd: user 'robin' already exists
useradd: user 'catwoman' already exists
useradd: user 'gordon' already exists
useradd: user 'joker' already exists
useradd: user 'penguin' already exists
player@playground:~$ sudo passwd batman
# Dann eingeben: darkknight
New password:
Retype new password:
passwd: password updated successfully
player@playground:~$ su - batman
# Passwort: darkknight
Password:
batman@playground:~$ █

```

```
GNU nano 7.2
#!/bin/bash

# Create groups first
groupadd heroesNN # Replace NN with 28
groupadd friends
groupadd police
groupadd villains

# Create directories
mkdir -p /home/heroesNN
mkdir -p /home/friends
mkdir -p /home/police
mkdir -p /home/villains

# Create users with passwords
useradd -m -c "YourName" -d /home/heroesNN/admin28 -s /bin/bash -g heroesNN admin28
useradd -m -c "Bruce Wayne" -d /home/heroesNN/batman -s /bin/bash -g heroesNN batman
useradd -m -c "Dick Grayson" -d /home/friends/robin -s /bin/bash -g friends robin
useradd -m -c "Selina Kyle" -d /home/friends/catwoman -s /bin/bash -g friends catwoman
useradd -m -c "James Gordon" -d /home/police/gordon -s /bin/bash -g police gordon
useradd -m -c "The Joker" -d /home/villains/joker -s /bin/bash -g villains joker
useradd -m -c "Oswald Cobblepot" -d /home/villains/penguin -s /bin/bash -g villains penguin

# Set passwords
echo "admin28:linux!is!c00l!" | chpasswd
echo "batman:darkknight" | chpasswd
echo "robin:nightwing" | chpasswd
echo "catwoman:meowmeow" | chpasswd
echo "gordon:law-n-order" | chpasswd
echo "joker:whysoserious" | chpasswd
echo "penguin:fish!" | chpasswd
```